Pronob Kumar Barman

Email 443-931-5511 LinkedIn GitHub Google Scholar Website

EDUCATION

University of Maryland, Baltimore County (UMBC) Expected Dec 2025
Ph.D. Information Systems | GPA: 3.85/4.00
Research Focus: Human-Centered Generative Models & Recommendation Systems

University of Kentucky May 2021
M.S. Statistics | GPA: 3.85/4.00

University of Dhaka Jan 2017
M.S. & B.S. Statistics | GPA: 3.46/4.00

PROFESSIONAL EXPERIENCE

PhD Researcher | UMBC Jan 2022–Present

- Developed a novel framework integrating AI with social science principles, resulting in a 30% increase in user engagement within online health communities.
- Validated recommendation algorithms through A/B testing, achieving 75% higher accuracy than industry-standard models and enhancing patient engagement metrics.
- Accelerated AI healthcare research by developing and open-sourcing two frameworks: (gSTM R package and gDMR Python codebase), reducing model development time by 40% while significantly increasing computational efficiency across multiple research domains.

Teaching Assistant | UMBC Jan 2022–Present

- Designed specialized workshops on advanced ML algorithms and LLM fine-tuning workflows, achieving student satisfaction scores of 4.8/5.0 across evaluations.
- Mentored 30+ graduate students in building scalable ML pipelines on AWS SageMaker and Google Vertex AI, resulting in 12 successful industry-ready projects.

Machine Learning Engineer | Technuf LLC Jun 2024–Aug 2024

- Built enterprise-grade RAG pipelines that processed 50,000+ daily SIEM security logs, transforming unstructured data into actionable threat intelligence.
- Optimized an anomaly detection system using statistical models and deep learning, reducing false positives by 20% and saving analysts 10 hours weekly.

Machine Learning Engineer | Technuf LLC Aug 2022–Dec 2022

- Enhanced data infrastructure through SQL query optimization for 200K+ healthcare records, slashing processing latency by 40% and enabling real-time analytics.
- Created unified Power BI visualizations from disparate healthcare data, empowering data-driven decisions across 15 departmental teams.

Deputy Director, Data Science | Bangladesh Bank Apr 2018–Dec 2021

- Spearheaded ML-based credit risk assessment models that reduced loan defaults by 10% across the national banking portfolio, preventing approximately $25M in losses.
- Led team of 5 data scientists in creating mission-critical financial intelligence dashboards used by central bank executives for monetary policy decisions.

TECHNICAL SKILLS

- **Languages & Frameworks:** Python, R, SQL, Java, PyTorch, TensorFlow, scikit-learn, Keras
- **AI & NLP:** GPT-3/4, LangChain, Hugging Face, BERT, SpaCy, NLTK, Transformers
- **Cloud & Data:** AWS SageMaker, GCP, Docker, Kubernetes, PostgreSQL, Pinecone, ChromaDB, Tableau, Power BI

SELECTED PROJECTS

NLP-Powered Customer Support System GitHub

- Implemented transformer models on AWS SageMaker to process customer queries in real-time, achieving 85% accuracy in intent classification and reducing response time by 40%.

ML Model Deployment for Digit Classification Github

- Constructed a Flask-based web application with TensorFlow on Google Cloud Run, enabling real-time predictions with 98% accuracy through a public API endpoint.

Medical Chatbot with Llama2 Github

- Designed a medical chatbot using Llama2 that accelerated patient interaction flow by 30% and achieved a patient satisfaction rating of 4.7/5 in post-interaction surveys.

Predictive Policing with GAN

- Formulated a Generative Adversarial Network model to mitigate algorithmic bias in predictive policing systems, reducing false-positive rates by 20% while maintaining detection accuracy.

PUBLICATIONS

- **Barman, P. K.**, Reynolds, T. L., & Foulds, J. (2025). "Facilitating Online Healthcare Support Group Formation using Topic Modeling." *Proceedings of the 19th World Congress on Medical and Health Informatics (MedInfo 2025)*.
- Rachael, K., Reynolds, T. L., Yus R. & **Barman, P. K.** (2024). "Acceptance of Occupancy Dashboards among Undergraduate Students: Towards Everyday Decision-making Infrastructure that can be Leveraged in Pandemics." *ACM Conference on Computer-Supported Cooperative Work and Social Computing*. (Under Review)
- Balogun, Z. A., **Barman, P. K.**, Onwumbiko, B. K., & Reynolds, T. L. (2024). "User-generated Data for Different Types of Mobile-Personal Health Records: A Computationally-guided Qualitative Analysis Approach." *AMIA Annual Symposium, 2024*.
- **Barman, P. K.** (2023). "Detecting Abnormal Health Conditions in Smart Home Using a Drone." *arXiv preprint arXiv:2310.05012*.

GRANTS & AWARDS

UMBC GSA Research Grant (2024): $1,000 for research on healthcare group formation models.